SI 140A-02 Probability & Statistics for EECS, Fall 2025
Homework 11

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Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1

(20pts) (Bounds on a Normal tail probability). Let $Z \sim \mathcal{N}(0,2)$. Following the Example 10.1.13 of the Textbook **BH**.
(a) Find the exact value of $P(|Z|>3)$ in terms of $\Phi(\cdot)$.
(b) Find the three approximations with upper bounds from Markov’s, Chebyshev’s, and Chernoff's inequalities, respectively. Which is the best approximation?

Solution

(a) Exact Value

Since $Z \sim \mathcal{N}(0,2)$, we have $\sigma = \sqrt{2}$. By standardizing:

$$
\begin{aligned}
P(|Z|>3) &= P(Z>3) + P(Z<-3) \\
&= 2P(Z>3) \quad \text{(by symmetry)} \\
&= 2P\left(\frac{Z}{\sqrt{2}} > \frac{3}{\sqrt{2}}\right) \\
&= 2\left(1-\Phi\left(\frac{3}{\sqrt{2}}\right)\right) \\
&= 2\Phi\left(-\frac{3}{\sqrt{2}}\right)
\end{aligned}
$$

Note: $\frac{3}{\sqrt{2}} \approx 2.12$.

(b) Approximations

1. Markov's Inequality We apply Markov's inequality to $|Z|$. First, we find $E[|Z|]$ for $Z \sim \mathcal{N}(0,\sigma^2)$:

$$E[|Z|] = \sqrt{\frac{2}{\pi}}\sigma = \sqrt{\frac{2}{\pi}} \cdot \sqrt{2} = \frac{2}{\sqrt{\pi}}$$

Then:

$$P(|Z|>3) \le \frac{E[|Z|]}{3} = \frac{2/\sqrt{\pi}}{3} = \frac{2}{3\sqrt{\pi}} \approx 0.376$$

2. Chebyshev's Inequality Using $Var(Z) = \sigma^2 = 2$:

$$P(|Z-0|>3) \le \frac{Var(Z)}{3^2} = \frac{2}{9} \approx 0.222$$

3. Chernoff's Inequality The MGF of $Z \sim \mathcal{N}(0,2)$ is $M_Z(t) = e^{\frac{1}{2}(2)t^2} = e^{t^2}$. For the one-sided tail $P(Z>3)$:

$$P(Z>3) \le e^{-3t}E[e^{tZ}] = e^{-3t}e^{t^2} = e^{t^2-3t}$$

To find the tightest bound, we minimize the exponent $g(t) = t^2 - 3t$.

$$g'(t) = 2t - 3 = 0 \implies t = 1.5$$

The minimum value is $e^{1.5^2-3(1.5)} = e^{2.25-4.5} = e^{-2.25}$. Considering two tails:

$$P(|Z|>3) \le 2e^{-2.25} \approx 2(0.1054) \approx 0.2108$$

Conclusion: Comparing the bounds:

- Markov: ≈ 0.376
- Chebyshev: ≈ 0.222
- Chernoff: ≈ 0.211

The **Chernoff bound** provides the best approximation.

Problem 2

(15pts) Let X and Y be i.i.d. continuous r.v.s. Assume that the various expressions below exist. Write the most appropriate of $\leq, \geq, =$, or ? in the blank for each part.

Solution

1. $E(\sqrt{X}) \leq \sqrt{E(X)}$

   Reasoning: The function $g(x) = \sqrt{x}$ is concave. By Jensen's Inequality ($E[g(X)] \leq g(E[X])$ for concave functions), the inequality holds.

2. $e^{-E(X)} \leq E(e^{-X})$

   Reasoning: The function $g(x) = e^{-x}$ is convex. By Jensen's Inequality ($E[g(X)] \geq g(E[X])$ for convex functions), the inequality holds.

3. $P(X > Y + 3) = P(Y > X + 3)$

   Reasoning: Since X and Y are i.i.d., the distribution of $X - Y$ is symmetric about 0. Thus $P(X - Y > 3) = P(Y - X > 3)$.

4. $P(X > Y + 3) \leq P(X > Y - 3)$

   Reasoning: The event $\{X > Y + 3\}$ implies $\{X > Y - 3\}$ (since $Y + 3 > Y - 3$). Therefore, the probability of the first is less than or equal to the second.

5. $E(X^4) \geq (E(XY))^2$

   Reasoning: Since X, Y are independent, $E(XY) = E(X)E(Y) = (E(X))^2$. Thus we compare $E(X^4)$ vs $(E(X))^4$. By Jensen's inequality on the convex function x^4, $E(X^4) \geq (E(X))^4$.

Problem 3

(15pts) Show that for every $c > 0$, $P(|\hat{p} - p| > c) \leq \frac{1}{4nc^2}$.

Solution

Let X_i be the indicator that the i-th person supports the policy. Then $\hat{p} = \frac{1}{n}\sum_{i=1}^{n} X_i$. The expectation and variance of $\hat{p}$ are:

$$E[\hat{p}] = p, \quad Var(\hat{p}) = \frac{p(1-p)}{n}$$

Applying Chebyshev's Inequality to $\hat{p}$:

$$P(|\hat{p} - E[\hat{p}]| > c) \leq \frac{Var(\hat{p})}{c^2} = \frac{p(1-p)}{nc^2}$$

To maximize the bound, we find the maximum of $f(p) = p(1-p)$. The maximum occurs at $p = 1/2$, where $p(1-p) = 1/4$. Thus:

$$P(|\hat{p} - p| > c) \leq \frac{1}{4nc^2}$$

Problem 4

(10pts) Let $X \sim \text{Expo}(\lambda)$. Find $E(X|X<1)$ in two different ways.

Solution

(a) By Calculus

The conditional PDF is $f_{X|X<1}(x) = \frac{f_X(x)}{P(X<1)} = \frac{\lambda e^{-\lambda x}}{1-e^{-\lambda}}$ for $0 < x < 1$.

$$\begin{aligned} E(X|X<1) &= \int_0^1 x \frac{\lambda e^{-\lambda x}}{1-e^{-\lambda}} dx \\ &= \frac{1}{1-e^{-\lambda}} \int_0^1 \lambda x e^{-\lambda x} dx \end{aligned}$$

Using integration by parts (let $u = \lambda x, dv = e^{-\lambda x}dx$ implies $du = \lambda dx, v = -\frac{1}{\lambda}e^{-\lambda x}$? No, simpler to let $t = \lambda x$):

$$\int_0^1 \lambda x e^{-\lambda x} dx = \frac{1}{\lambda} \int_0^\lambda t e^{-t} dt$$

We know $\int te^{-t}dt = -(t+1)e^{-t}$.

$$\int_0^\lambda te^{-t}dt = [-(t+1)e^{-t}]_0^\lambda = -(\lambda+1)e^{-\lambda} - (-1) = 1 - (\lambda+1)e^{-\lambda}$$

Substituting back:

$$\begin{aligned} E(X|X<1) &= \frac{1}{\lambda(1-e^{-\lambda})}(1 - \lambda e^{-\lambda} - e^{-\lambda}) \\ &= \frac{1}{\lambda} - \frac{e^{-\lambda}}{1-e^{-\lambda}} = \frac{1}{\lambda} + \frac{e^{-\lambda}}{e^{-\lambda}-1} \end{aligned}$$

(b) Law of Total Expectation

$$E[X] = E[X|X<1]P(X<1) + E[X|X \geq 1]P(X \geq 1)$$

We know $E[X] = 1/\lambda$, $P(X \geq 1) = e^{-\lambda}$, and $P(X < 1) = 1 - e^{-\lambda}$. By the memoryless property, $E[X|X \geq 1] = 1 + E[X] = 1 + 1/\lambda$.

$$\begin{aligned} \frac{1}{\lambda} &= E[X|X<1](1-e^{-\lambda}) + \left(1 + \frac{1}{\lambda}\right) e^{-\lambda} \\ \frac{1}{\lambda} &= E[X|X<1](1-e^{-\lambda}) + e^{-\lambda} + \frac{1}{\lambda}e^{-\lambda} \\ \frac{1}{\lambda}(1-e^{-\lambda}) - e^{-\lambda} &= E[X|X<1](1-e^{-\lambda}) \end{aligned}$$

Dividing by $(1-e^{-\lambda})$:

$$E[X|X<1] = \frac{1}{\lambda} - \frac{e^{-\lambda}}{1-e^{-\lambda}}$$

Problem 5

(20pts) Two envelopes with Unif(0, 1) money. Strategy: observe U, stick with prob U, switch with prob $1-U$.

Solution

Let $X_1, X_2 \sim \text{Unif}(0,1)$ be independent. We observe X_1. Stick probability = X_1, Switch probability = $1 - X_1$.

(a) Probability of getting the larger amount

We win if: 1. $X_1 > X_2$ and we stick. 2. $X_1 < X_2$ and we switch.

$$
\begin{aligned}
P(\text{Win}) &= \int_0^1 \int_0^1 P(\text{Win}|X_1 = x_1, X_2 = x_2) dx_2 dx_1 \\
&= \iint_{x_1 > x_2} x_1 dx_2 dx_1 + \iint_{x_1 < x_2} (1 - x_1) dx_2 dx_1 \\
&= \int_0^1 \left(\int_0^{x_1} x_1 dx_2 \right) dx_1 + \int_0^1 \left(\int_{x_1}^1 (1 - x_1) dx_2 \right) dx_1 \\
&= \int_0^1 x_1^2 dx_1 + \int_0^1 (1 - x_1)^2 dx_1 \\
&= \left[\frac{x_1^3}{3} \right]_0^1 + \left[-\frac{(1 - x_1)^3}{3} \right]_0^1 \\
&= \frac{1}{3} + \frac{1}{3} = \frac{2}{3}
\end{aligned}
$$

(b) Expected value of received amount

Let R be the reward. Given X_1, X_2, the expected reward is $X_1(X_1) + X_2(1 - X_1)$.

$$
\begin{aligned}
E[R] &= \int_0^1 \int_0^1 [x_1^2 + x_2(1 - x_1)] dx_2 dx_1 \\
&= \int_0^1 \left[x_1^2 x_2 + \frac{x_2^2}{2}(1 - x_1) \right]_{x_2=0}^1 dx_1 \\
&= \int_0^1 \left(x_1^2 + \frac{1}{2}(1 - x_1) \right) dx_1 \\
&= \int_0^1 \left(x_1^2 + \frac{1}{2} - \frac{x_1}{2} \right) dx_1 \\
&= \frac{1}{3} + \frac{1}{2} - \frac{1}{4} = \frac{4 + 6 - 3}{12} = \frac{7}{12}
\end{aligned}
$$

Problem 6

(20pts) 21 women, 14 men (35 total). Each has disease with prob p. Found exactly 5 have disease (S). Find expected men/women with disease given S.

Solution

Let I_j be the indicator that person j has the disease. We are given the event $S : \sum_{j=1}^{35} I_j = 5$. Due to symmetry, for any person j, $P(I_j = 1|S)$ is the same.

$$E\left[\sum_{j=1}^{35} I_j \middle| S\right] = 5 \implies \sum_{j=1}^{35} E[I_j|S] = 5 \implies 35E[I_1|S] = 5$$

Thus, $P(I_j = 1|S) = \frac{5}{35} = \frac{1}{7}$.

Expected number of men with disease: Let M be the number of men with the disease. There are 14 men.

$$E[M|S] = \sum_{j\in\text{Men}} E[I_j|S] = 14 \times \frac{1}{7} = 2$$

Expected number of women with disease: Let W be the number of women with the disease. There are 21 women.

$$E[W|S] = \sum_{j\in\text{Women}} E[I_j|S] = 21 \times \frac{1}{7} = 3$$